Fractions, Decimals & Percentages

PAPER B — MEDIUM

Name ____________________ **Date** ____________ **Score** ____________

Before you start

- No calculator. Every mark here is available with pencil, paper and the nine equivalents you memorised.
- Convert everything into **decimals** when you need to compare or order — it is almost always the fastest route.
- For a percentage of an amount, build it from 10%, 5% and 1%, or use a single multiplier. Pick whichever the numbers prefer.
- Write the unit and, where the question asks for a change, the word **increase** or **decrease**. Those words carry marks.

SECTION A — SHORT ANSWERS

1 Write each of these as a percentage. [3]

(a) 0.08 [1]

(b) 1/4 [1]

(c) 7/20 [1]

2 Write 65% as a fraction in its simplest form. [1]

Answer

3 Work out 2/3 + 1/6, giving your answer in its simplest form. [2]

Answer

4 Find 15% of 260 without a calculator. Show the building blocks you used. [2]

Answer

5 Write these three numbers in order, smallest first. You must show your working. [2]

0.45 2/5 43%

Answer

6 Work out 4/5 ÷ 2/3, giving your answer as a mixed number. [2]

Answer

SECTION B — PROBLEMS

7 Increase 240 by 35%. [2]

Answer

8 Decrease 180 by 20%. [2]

Answer

9 A theatre has 600 tickets for a show. 45% are sold on the first day, and 1/4 of the original 600 are sold on the second day. [3]

(a) How many tickets are sold altogether over the two days? [2]

(b) How many tickets remain unsold? [1]

10 Write 0.375 as a fraction in its simplest form. [2]

Answer

11 The price of a book falls from ₹250 to ₹200. Calculate the percentage decrease. [2]

Answer %

12 Work out 1 2/5 × 1 1/4, giving your answer as a mixed number in its simplest form. [3]

Answer

13 A recipe uses 3/4 kg of sugar to make 12 biscuits. How much sugar is needed for 20 biscuits? Give your answer in grams. [2]

Answer g

14 Which is larger, 5/8 or 0.6? You must justify your answer with a calculation. [2]

Answer